

A skilled professional in wide range of current technical landscape in industries such as Retail, Healthcare, GIS Location Systems, Media and Information Services, Telecom etc. Possess 12 years of deep experience in AI/ML and Big Data, creating strategies, organising and negotiating, designing architecture, coaching and making quick decisions through proven experience of automation and translating business vision into initiatives that improve performance, profitability, training, development, and growth.

Extensive hands on versatile technical skills in *Python, PySpark, Pandas, Gen AI(LLM), Langchain, NLP, OCR, RAG, LightGBM, TensorFlow, Snowflake, Airflow, Ray, Node.js, Neo4J(Graph), Athena, SQL, Vector and NoSQL databases* etc.

Technical Skills:

- Programming Languages : Python, Node.js, Java, Javascript
- AI/ML and Data Analytics : Gen AI(LLM), Langchain, Pandas, LightGBM, RAG, Glue, TensorFlow, MLflow, Kibana, Sagemaker, Scikit-Learn, Vertex AI, Redshift
- Big Data & Backend Frameworks : PySpark, Kafka, Flink, Hadoop, Hive, Dask, Ray, Airflow, Snowflake, Django, FastAPI
- Scripting Languages : Shell/Bash
- Databases : PostgreSQL, MySQL, ElasticSearch, HBase, MongoDB, DynamoDB, Neo4J, Athena, VectorDB, Pinecone
- Cloud Technologies : AWS, GCP, Azure
- Version Control/CICD : Git, Docker, Kubernetes, Jenkins, Terraform, AWS Cloudformation, Github Actions
- Developer Tools : IDEs, JIRA, Confluence

Professional Experience:

Sivisoft/SquareML | June 2021 – Current

Worked as Lead Data Scientist with Sivisoft/SquareML for clients as noted below.

Womba (Medical Data Aggregator), Parameds (PDC) and Denver Health:

Design and development of GenAI based data processing platform enabling underwriters to summarise unstructured content in the PDFs using custom LLM. This automated and enhanced the underwriter's efficiency by 30 times

The technology stack comprised of Python, MySQL, Django, AWS Step Functions, Sagemaker, Batch, Lambda, S3, LLM, Vertex AI, Langchain, TensorFlow, Docker, ElasticSearch, Pandas, PySpark, NLP, OCR, RAG

VRTAC-QM Chatbot (SDSU University):

Design and development of GenAI based chatbot to enhance user experience by providing instant, relevant information and ability to find information linked to the site efficiently.

The technology stack comprised of Python, PostgreSQL, FastAPI, AWS Batch, S3, OpenAI LLM, Langchain, Docker, Pinecone, RAG, Drupal

EPP (Early Purchase Prediction) and ARF (Automated Retraining Framework):

Lead the design and development of the real-time streaming data platform, enabling client specific AI/ML use-cases, in-session predictions, automated model retraining, and executing marketing campaigns to drive revenue up to 10x. This was done for clients such as Wal-Mart, BestBuy, Advance Auto Parts, Verizon, etc

The technology stack comprised of Python, Kafka, HBase, MLflow, LGBM, TensorFlow, FastAPI, S3.

Senior Software Engineer | HERE Technologies | Sep 2015 - April 2021

Worked on various projects for HERE Technologies as described below.

- HERE Platform - Developed the batch and real-time streaming data pipelines support for Python in the #1 ranked HERE Platform to process terabytes of data and millions of events per day.
Clients: Ford, Daimler. Tech Stack: Python, Ray, Dask, Spark, Kubernetes
- MLflow Plugin - Lead the design and development of the HERE MLflow plugin, to manage Machine Learning (ML) experiments on the platform and share with others or make it available on the HERE Marketplace. The technology stack comprised of Python and MLflow
- Platform Python SDK - Crafted the HERE Data SDK for Python, a well-known open-source set of analytics and data engineering tools that have been modified to work with the HERE platform.
Clients: Ford, BMW.
- Road Attributes Detection - Developed a custom object detection YOLO-based model to detect the basic road attributes like vehicles, traffic signals, sign boards, etc with 80%+ accuracy.
Tech Stack: Python, PyTorch
- DataHub/XYZ - Involved in the development of the HERE Data Hub, a cloud service designed to make map data access simple and scalable to process millions of events per day.
Tech Stack: Node.js, Java, AWS Lambda, S3, API Gateway, Postgres, DynamoDB, Elasticsearch, Spark
- TQS Score - Developed a regression-based predictive machine learning

model to score the quality of places data based on supplier rating, freshness, etc.

Tech Stack: Python, Scikit-learn, Hive, HBase

- Places Data Store (PDS) - Developed an efficient system, PDS, to receive, store, manage and deliver high-quality scalable places data through the Places Manager API, Product Builder. This system helps process 1.2 billion plus places' records in a day.

Clients: Facebook, Yelp, etc.

Tech Stack: Java, MapReduce, HBase, Hive, Oozie, Pig

Systems Engineer | Tata Consultancy Services | Dec 2012- Sep 2015

Worked with Tata Consultancy Services in the capacity of Systems Engineer performing various tasks as described below.

- MCube - A second-screen iOS application for the United States market regarding media details, schedules, and recommendations based on their activities. Designed and developed a data pipeline to fetch data from multiple sources for a collaborative filtering-based recommendation engine to recommend users' favourite media content based on their social profile and clickstream analysis implemented with graph based Neo4J database.

Tech Stack: Java, AWS EMR, S3, Hive, Neo4j

Education:

Birla Institute of Technology and Science (BITS), Pilani - 2018 – 2020

Post Graduate Program in Artificial Intelligence and Machine Learning

Welingkar Institute of Management / PGDM, Mumbai - 2016 - 2018

Post Graduate Diploma in Management - IT Projects Management (PGDM)

University of Mumbai / B.E., Mumbai - 2008 - 2012

Bachelor of Engineering - Computer Engineering (B.E.)

Certificate and Trainings:

Leadership Accelerator Servant Leadership Program, HERE Technologies | 2021

MongoDB for Java Developers Certification, MongoDB | 2015

Google Analytics Certification, Google | 2014

Oracle Certified Professional Java Programmer, Oracle | 2012